ADAPT-BNS

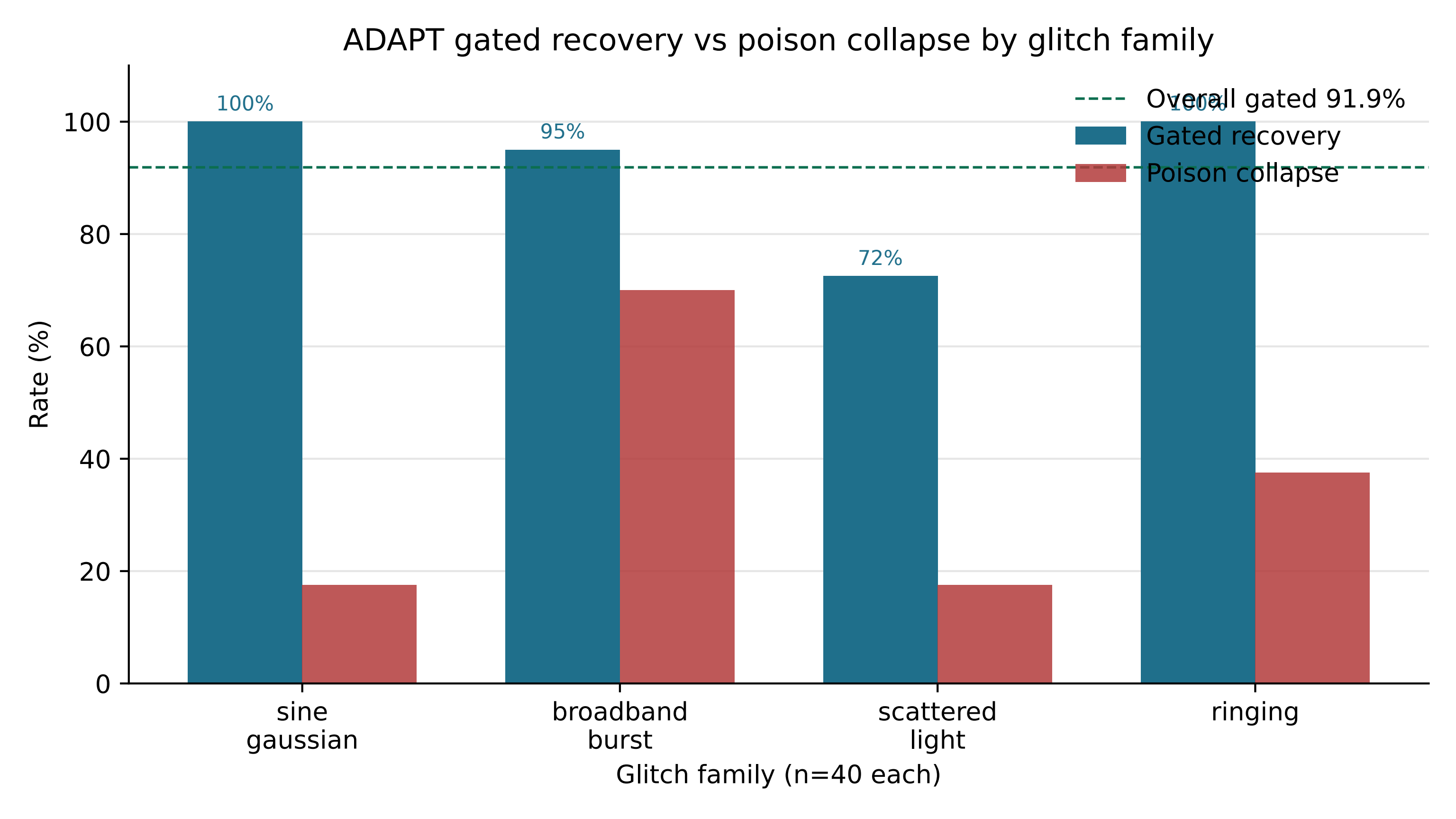
Synthetic injection stress test · detect-and-gate + frozen DINGO

GATED RECOVERY **91.9%** 160 glitch cells	**CLEAN KAT PASS** **87%** 20/23 draws
DETECTOR FIRE **100%** gates on glitchy H1	**POISON COLLAPSE** **36%** rail criterion
MEAN |Δ MED D_L| **1.38 Mpc** among successes	**HELD-OUT RECOVERY** **100%** ringing family

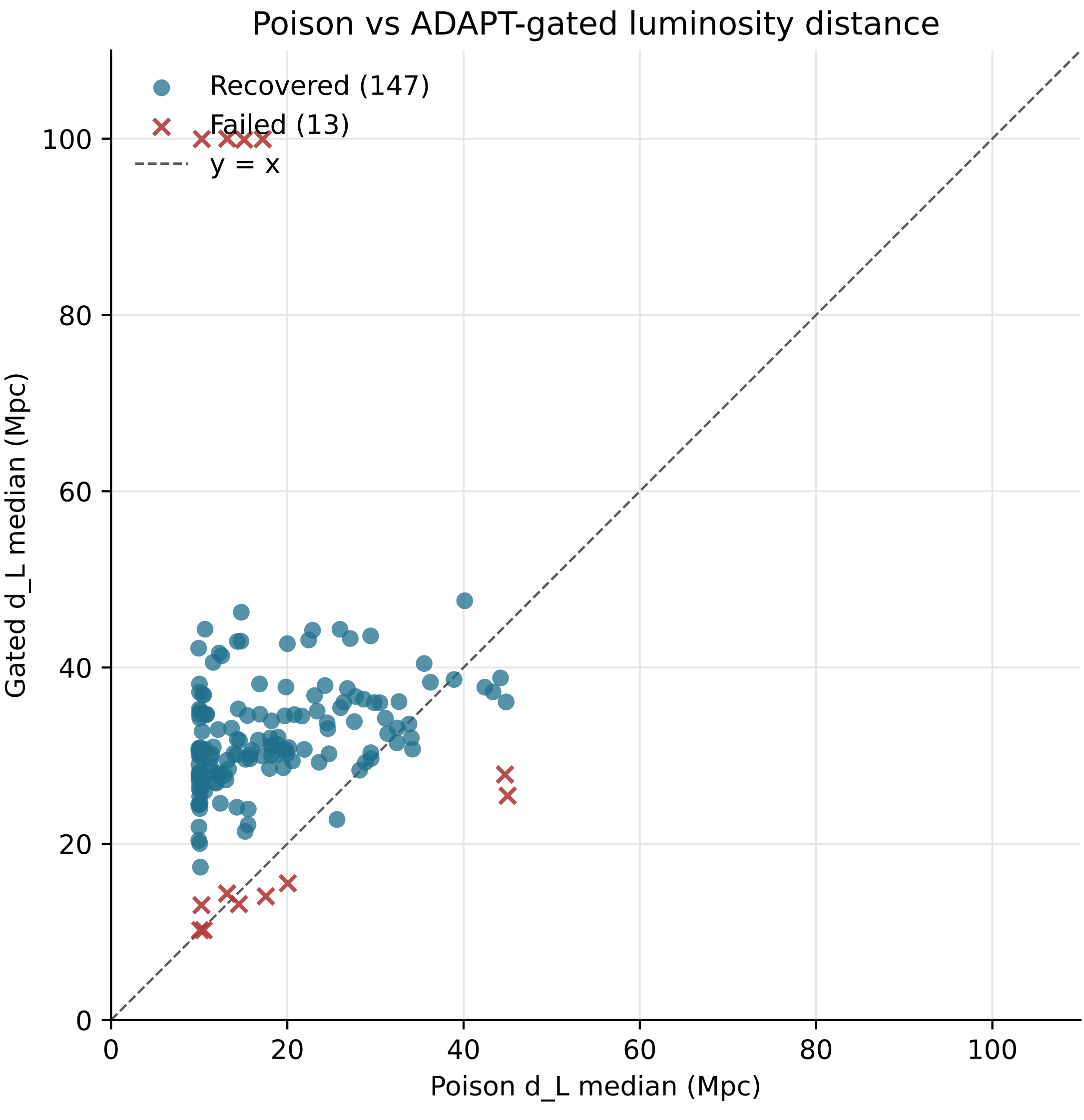
Takeaways

- PE engine: frozen official GW170817 DINGO (no NSF retrain).
- ADAPT path: spectrogram detector → Tukey gate → matched-delta FD rebuild → original ASD.
- 20 KAT-pass synthetic BNS injections × 8 glitch cells (SG, burst, scattered light, ringing).
- Recovery = gated d_L CI overlaps clean and |med − clean med| ≤ 10 Mpc.
- Weak spot: scattered_light (72.5%); held-out ringing = 100%.

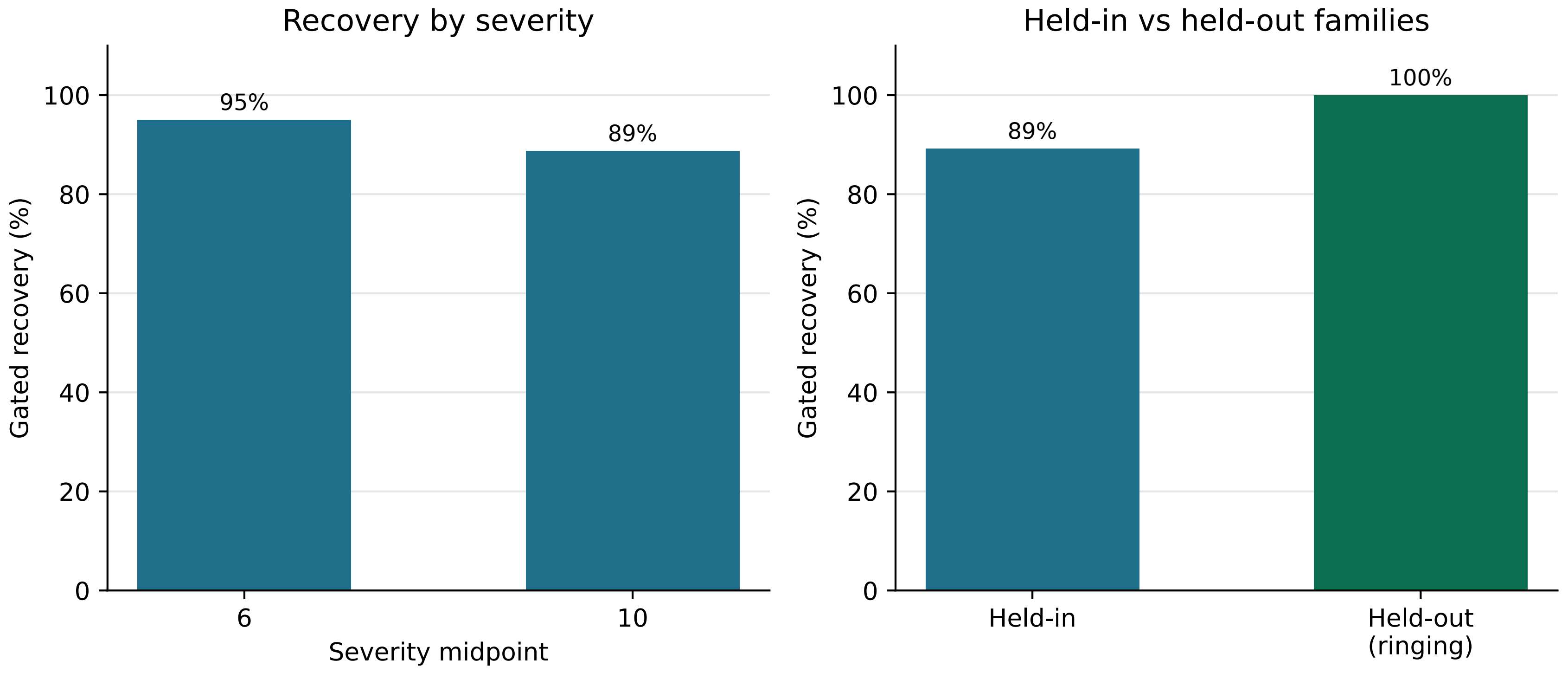
Source: results/stress_test_synthetic_bns_v1 · N=512 NN samples · vector PDF (TrueType)



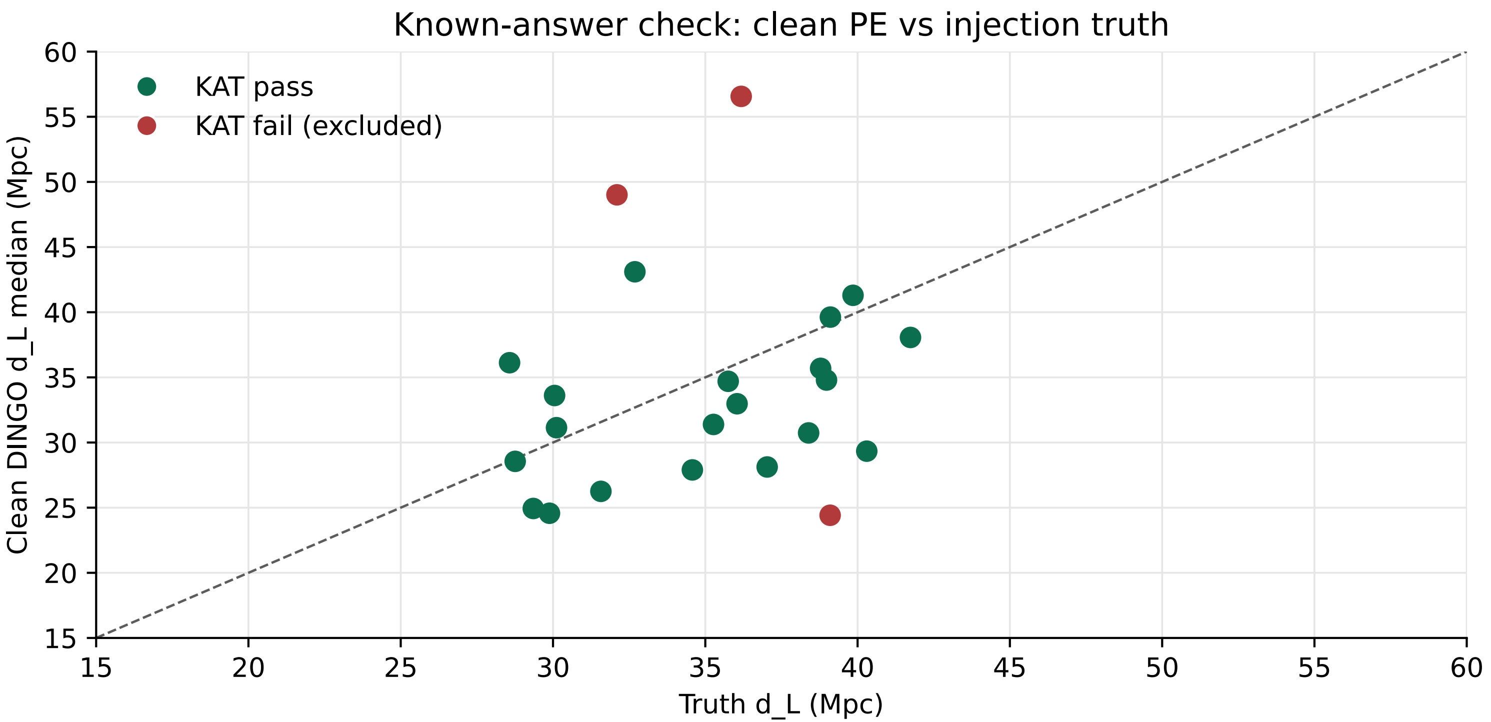
ADAPT gated recovery vs poison collapse by glitch family
Overall gated 91.9%
Gated recovery
Poison collapse
100%
95%
72%
100%
Rate (%)
0
20
40
60
80
100
sine gaussian
broadband burst
scattered light
ringing
Glitch family (n=40 each)



Poison vs ADAPT-gated luminosity distance
Recovered (147)
Failed (13)
y = x
Gated d_L median (Mpc)
Poison d_L median (Mpc)
0
20
40
60
80
100



Recovery by severity
Gated recovery (%)
0
20
40
60
80
100
95%
89%
6
10
Severity midpoint
Held-in vs held-out families
Gated recovery (%)
0
20
40
60
80
100
89%
100%
Held-in
Held-out
(ringing)


Known-answer check: clean PE vs injection truth
KAT pass
KAT fail (excluded)
Clean DINGO d_L median (Mpc)
Truth d_L (Mpc)
15
20
25
30
35
40
45
50
55
60

Methods (brief)

1. Synthetic CBC injections use build_base_domain_injection with GW170817 ASDs and fixed sky.
2. Per-event chirp_mass_proxy is set from truth for GNPE heterodyning.
3. KAT: |med chirp_mass − truth| ≤ 0.05 and |med d_L − truth| ≤ 12 Mpc.
4. Glitch panel: sine_gaussian, broadband_burst, scattered_light, ringing × severity {6,10} × Welch.
5. Poison = glitchy FD + Welch ASD; ADAPT = detect → Tukey ±0.4 s → matched_delta → original ASD.
6. Frozen DINGO weights only; detector checkpoint: glitch_detector_v1.
7. Reproduce: scripts/stress_test_synthetic_bns.py (see REPRODUCE.md).

Artifact index

```
ADAPT_synthetic_bns_report.pdf  — this report
figures/*.pdf / *.svg           — individual vector plots
summary.json / results.csv      — numeric source of truth
failures.csv                    — gated non-recoveries
```